

Q161: Korff's **1em:E** is a typographical inversion,
and my localised rim-hook weight is his prob on the nose

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Abstract

The connection “*Korff’s Hecke rim-hook weight is local in Petrov’s coordinates*” (DREAM 2026-09-17 c2) was left conditional on one discrepancy: my localised weight $t^{\ell-1}(t-1)(-1/t)^{r-1}$ disagreed with the operator weight $(t^{-1}-1)(-t)^{r-1}$ printed in **1em:E** of Korff [arXiv:1906.02565](#), by $t^{\pm 2(r-1)}$ and an apparently missing $t^{\ell-1}$. Two readings were on the table: an inverted- t convention, or matrix-element-versus-character-weight with a grading shift. **Both are wrong, and there is no mathematical gap.** The printed statement of **1em:E** contradicts Korff’s own definitions, his own proof of that lemma, and his own consistency condition $\text{prob}(b) = (t-1) \text{wt}(b)$; all three give the same corrected formula, and against that formula my localisation agrees exactly, the residual t^ℓ being the prefactor $(xt)^{j-i}$ that his half-vertex operator attaches to each E_{ij} .

1 The two formulas

Korff [arXiv:1906.02565](#) (source file **draft_02.tex**). For a broken rim hook b with connected components h , each occupying $r(h)$ rows and $c(h)$ columns (eq. **Hecke_wt**, source l. 462):

$$\text{wt}(b) = (t-1)^{\#b-1} \prod_{h \in b} (-1)^{r(h)-1} t^{c(h)-1}, \quad (1)$$

and (l. 1171) the six-vertex normalisation $\text{prob}(b) = (t-1) \text{wt}(b)$. The deformed fermions (**C12Ft+**, **C12Ft-**, source ll. 574–590) are

$$\psi_i^+(t)\sigma = \begin{cases} (\sigma + \epsilon_i) \prod_{j>i} (-t)^{\sigma_j} & \sigma_i = 0 \\ 0 & \sigma_i = 1 \end{cases} \quad \psi_i^-(t)\sigma = \begin{cases} (\sigma - \epsilon_i) \prod_{j>i} (-t)^{-\sigma_j} & \sigma_i = 1 \\ 0 & \sigma_i = 0, \end{cases}$$

and $E_{ij}(t) := (1-t)\psi_i^-(t)\psi_j^+(t)$. **1em:E** (ll. 626–635) states: for $i < j$ and λ/μ a connected rim hook of length $\ell = j - i$ occupying r rows,

$$E_{ij}(t) \sigma(\mu, c) = (t^{-1} - 1)(-t)^{r-1} \sigma(\lambda, c). \quad (2)$$

My localisation predicts, for the same move, $t^{\ell-1}(t-1)(-1/t)^{r-1}$.

2 The corrected lemma

Lemma 1. *For $i < j$, $\sigma_i = 1$, $\sigma_j = 0$, let $m = \#\{l : i < l < j, \sigma_l = 1\}$, so that the rim hook added occupies $r = m + 1$ rows. Then, directly from **C12Ft±**,*

$$E_{ij}(t) \sigma(\mu, c) = (1 - t^{-1}) (-t)^{1-r} \sigma(\lambda, c).$$

Proof. Write $n_{>k} = \sum_{l>k} \sigma_l$. Applying $\psi_j^+(t)$ first gives the factor $(-t)^{n_{>j}}$ and the diagram $\sigma' = \sigma + \epsilon_j$. In σ' the occupied positions above i are the m positions strictly between i and j , the position j itself, and the $n_{>j}$ positions above j ; so $\psi_i^-(t)$ contributes $(-t)^{-(m+1+n_{>j})}$. The two factors multiply to $(-t)^{-(m+1)}$, and

$$E_{ij}(t) = (1-t)(-t)^{-(m+1)} = \frac{1-t}{-t} (-t)^{-m} = (1-t^{-1})(-t)^{-m},$$

which is the claim since $m = r - 1$. $\square$

Note the two differences from (2): an overall sign, and the sign of the exponent of $(-t)$.

3 Three independent confirmations

(i) Korff's definitions, by exhaustive evaluation. Evaluating $(1-t)\psi_i^-(t)\psi_j^+(t)$ on Maya diagrams in a window $[-7, 7]$, over all partitions μ with at most three parts of size ≤ 5 and all admissible (i, j) : **2716** nonzero matrix elements. All 2716 agree with Lemma 1. **None** agrees with (2). The same sweep was run under all four plausible convention variants ($\prod_{j>i}$ against $\prod_{j<i}$; ψ^+ applied first against ψ^- first): (2) is reproduced by *none* of the four, so the disagreement is not a misreading of a convention.

(ii) Korff's own proof of lem:E. Source l. 639: "For each 1-letter between positions i and j ... there is a power of $-t^{-1}$." There are $m = r - 1$ such letters, giving $(-t^{-1})^{r-1} = (-t)^{1-r}$ — the corrected exponent. The proof states the lemma correctly; only the display inverts it.

(iii) Korff's own consistency condition — decisive. In the half-vertex operator (eq. HeckeA, l. 606)

$$A(x; t) = 1 + \sum_{r>0} \sum_{i_1 < j_1 < \dots < i_r < j_r} (xt)^{j_1 - i_1 + \dots + j_r - i_r} E_{i_1 j_1}(t) \dots E_{i_r j_r}(t),$$

every E_{ij} carries the prefactor $(xt)^{j-i}$, and Prop. prop:BFHecke / eq. BFA requires the single- E term to reproduce $\text{prob}(b) = (t-1) \text{wt}(b)$ for a connected hook. Hence $(xt)^\ell E_{ij} = x^\ell (t-1) \text{wt}(b)$, i.e.

$$E_{ij}(t) = t^{-\ell} (t-1) (-1)^{r-1} t^{c-1} = (t-1) (-1)^{r-1} t^{-r} = (1-t^{-1})(-t)^{1-r},$$

using $r + c - 1 = \ell$. Verified symbolically for every (ℓ, r) with $\ell \leq 8$: the corrected form satisfies Korff's consistency condition for all of them, the printed form for none.

4 The consequence: my localisation is Korff's prob, exactly

Proposition 2. *For a connected rim hook of length ℓ occupying r rows,*

$$\underbrace{t^{\ell-1} (t-1) (-1/t)^{r-1}}_{\text{my localised weight}} = (t-1) \text{wt}(b) = \text{prob}(b), \quad \text{and} \quad \text{my localised weight} = t^\ell E_{ij}(t).$$

Both identities hold for every (ℓ, r) with $1 \leq r \leq \ell$; verified symbolically on all 36 pairs with $\ell \leq 8$, and the second on all 2716 matrix elements of §3(i).

Proof. With $c = \ell + 1 - r$ we have $t^{\ell-1} \cdot t^{-(r-1)} = t^{\ell-r} = t^{c-1}$, so $t^{\ell-1} (t-1) (-1/t)^{r-1} = (t-1) (-1)^{r-1} t^{c-1} = (t-1) \text{wt}(b)$ by (1) with $\#b = 1$. The second identity is then $t^\ell (1-t^{-1})(-t)^{1-r} = (t-1) (-1)^{r-1} t^{\ell-r}$, which is the same expression. $\square$

So the unexplained prefactor $(t - 1)$ I had been carrying is the very factor Korff divides out at l. 1171, and the “missing” $t^{\ell-1}$ was never missing: a matrix element does not carry the grading that its own generating function supplies.

Remark 3 (why reading (i) looked plausible). The printed weight $(t^{-1} - 1)(-t)^{r-1}$ is not a $t \mapsto t^{-1}$ substitution of wt — it is wt of the *transposed* hook, up to $(-1)^\ell(t - 1)/t$, since interchanging $r \leftrightarrow c = \ell + 1 - r$ carries $(-1)^{r-1}t^{c-1}$ to $(-1)^{c-1}t^{r-1}$. A typo that lands on a genuine involution of the objects is exactly the kind that survives a plausibility check; the lesson is that the surrounding *proof* decided in one line what three readings of the *statement* could not.

Remark 4 (what this buys). In the occupancy coordinates of the commuting-ribbon paper, $r - 1 = |u|$ and $c - 1 = \ell - 1 - |u|$, so wt normalised as prob reads $\sigma_{\text{Korff}}(u) = (t - 1)t^{\ell-1}(t^{-1})^{|u|}$: a *height* weight of geometric type $AB^{|u|}$. Any weight assembled multiplicatively from per-letter data has this form, and by the height-weight corollary of that paper such weights commute across ranks only when $B = 1$ — Murnaghan–Nakayama. This is recorded there as **rem:local**. The identification of the class $AB^{|u|}$ with “realisable by a local six-vertex Boltzmann weight” rests on Petrov [arXiv:2609.18502](#) Rmk. 2.9, which is held at *agent-summary* depth only and is therefore **not** claimed here as proved.